Feedback Questionnaire

HiLDE prototype — hierarchical subspace exploration v1.3

Date:

No name — responses are reported only in aggregate or as anonymous quotes.

Instructions. For statements with a scale, mark *one* box: **1** = strongly disagree, **2** = disagree, **3** = neutral, **4** = agree, **5** = strongly agree. If you feel you cannot judge a statement from this session, mark **n/j** instead. Short written answers can be keywords — full sentences are not required.

Part 1 Background · before the demo

P1. Current role and field:

P2. Years of experience in data analysis: ☐ < 1 ☐ 1–3 ☐ 4–10 ☐ > 10

P3. How often do you work with tabular data with more than ~ 10 features?
☐ never ☐ a few times a year ☐ monthly ☐ weekly ☐ daily

P4. Rate your familiarity (1 = none, 5 = expert):

	1	2	3	4	5
a) Dimensionality reduction (PCA, UMAP, t-SNE)	☐	☐	☐	☐	☐
b) Clustering methods (e.g. k-means, DBSCAN)	☐	☐	☐	☐	☐
c) Reading scatter plots of projected data	☐	☐	☐	☐	☐

P5. Which tools do you mainly use when exploring a new dataset?

Part 2 The problem

	1	2	3	4	5	n/j
A1. Identifying subgroups of similar rows in a dataset with many features — and describing what makes them different — is a task that occurs in my own work.	☐	☐	☐	☐	☐	☐

A2. The last time this task (or a similar one) came up: how did you approach it?

Part 3 The warm-up dataset · handwritten digits

At the start of the session you explored the digits dataset yourself and then looked at one group more closely, descending into it. Every row there could be checked as a picture. The following statements refer to that part of the session.

	1	2	3	4	5	n/j
F1. The tool's displays (projection, score tiles, bars) would have let me judge what was in the group I examined, even without clicking any row images.	☐	☐	☐	☐	☐	☐
F2. Descending one level made the composition of that group clearer than the top-level view alone.	☐	☐	☐	☐	☐	☐
F3. To understand that group, I relied mainly on the row images rather than the tool's other displays.	☐	☐	☐	☐	☐	☐
F4. Some of the points the tool set aside as unassigned (grey) looked to me like they belonged to one of the groups.	☐	☐	☐	☐	☐	☐

F5. On the digits data you could check the tool's grouping directly, because every row is a picture. On most real datasets no such check exists. Does that change how far you would trust the tool's groups on data without such a check? Why?

Part 4 Finding structure · the wine dataset

	1	2	3	4	5	n/j
B1. The step-by-step drill-down showed me structure that I did not see in the first, global view.	☐	☐	☐	☐	☐	☐
B2. The regions the tool produced correspond to groups I would consider meaningful in the data.	☐	☐	☐	☐	☐	☐
B3. During drill-down, I lost track of which part of the data I was looking at.	☐	☐	☐	☐	☐	☐
B4. The per-region view (cluster projection, characteristics bars) gave me enough information to decide where to go next.	☐	☐	☐	☐	☐	☐

B5. Was there a moment where the tool's grouping surprised you — positively or negatively? What happened?

Part 5 Reading the interface

	1	2	3	4	5	n/j
C1. The overview projection (coloured regions, grey crosses for unassigned points) was easy to interpret.	☐	☐	☐	☐	☐	☐
C2. Selecting points in the region scatter plot worked the way I expected.	☐	☐	☐	☐	☐	☐
C3. The range chart made it easy to see which feature ranges describe my selection.	☐	☐	☐	☐	☐	☐
C4. The interface showed more information than I could make use of.	☐	☐	☐	☐	☐	☐

C5. Which single view was least clear to you, and why?

Part 6 The two descriptions · the wine dataset

You were shown two descriptions of one set of selected points, labelled Description A and Description B. Each also states how many of the selected points it covers. The following questions refer to those two descriptions as you saw them.

	1	2	3	4	5	n/j
D1. Description A was easy to read and understand.	☐	☐	☐	☐	☐	☐
D2. Description B was easy to read and understand.	☐	☐	☐	☐	☐	☐
D3. Based on Description A alone, I could tell a colleague which points are meant.	☐	☐	☐	☐	☐	☐
D4. Based on Description B alone, I could tell a colleague which points are meant.	☐	☐	☐	☐	☐	☐
D5. I would trust Description A to still apply if similar new data points were added.	☐	☐	☐	☐	☐	☐
D6. I would trust Description B to still apply if similar new data points were added.	☐	☐	☐	☐	☐	☐

D7. If you could report only one of the two to a colleague, which would you choose?

☐ Description A ☐ Description B ☐ No preference

Why?

	1	2	3	4	5	n/j
D8a. If the selection was changed during the session: the way *Description A* changed matched my expectation.	☐	☐	☐	☐	☐	☐
D8b. If the selection was changed during the session: the way *Description B* changed matched my expectation.	☐	☐	☐	☐	☐	☐

D9. What, if anything, would you change about how the descriptions are presented?

Part 7 Value and adoption

	1	2	3	4	5	n/j
E1. There is a concrete dataset in my own work on which I would want to try this tool.	☐	☐	☐	☐	☐	☐

If you agree: what kind of data?

	1	2	3	4	5	n/j
E2. Compared to my current approach (see A2), this workflow would save me effort when looking for subgroups.	☐	☐	☐	☐	☐	☐

E3. What is the strongest reason you would *not* adopt this tool in your work? (Please answer — critical answers are the most useful ones.)

E4. What single change would most increase the tool's value for you?

Thank you. Please hand the questionnaire back — the remaining questions are verbal.